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CN-CELEB: A CHALLENGING CHINESE SPEAKER RECOGNITION DATASET

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Abstract

Recently, researchers have set a more ambitious goal to tackle the speaker recognition task in unconstrained conditions, i.e., recording with any variation of ambient, channel, emotion, etc. However, most publicly available datasets are collected under constrained environments, i.e., with little noise and limited channel variation. These datasets tend to deliver over optimistic performance and cannot meet the request of designing robust algorithms that are suitable for real-life scenarios.

In this report, we present *CN-Celeb*, a large-scale speaker recognition dataset collected 'in the wild'. This dataset contains more than $130k$ utterances from $1,000$ Chinese celebrities, and covers 11 different conditions in real world. Experiments conducted with several state-of-the-art speaker recognition methods show that this *CN-Celeb* is a challenging dataset, and in real-life conditions, the performance of existing techniques might be much worse than it was thought.

Keywords: speaker recognition

1 Introduction

This report is an extension of our previous work [**?**]. Firstly, we will show the performance on the SOTA speaker recognition methods in this real-life *CN-Celeb* dataset. Secondly, we will give further analysis on *CN-Celeb*, and focus on several notable factors, including genres, duration, gender, age.

2 Dataset

Two datasets were used in our experiments: *VoxCeleb* and *CN-Celeb*.

VoxCeleb: A large-scale free speaker dataset collected by University of Oxford, UK [**?**] [**?**]. The entire dataset involves two parts: *VoxCeleb1* and *VoxCeleb2*. Data purging was conducted to remove the overlap between *VoxCeleb1* and *SITW* [**?**]. Finally, *VoxCeleb* is separated into two parts. The main part involves $1,236,567$ utterances from $7,185$ speakers, which is regarded as the training set. The rest part, which is the same as *SITW*, is regarded as the evaluation set. In our experiments, the *Eval. Core* in *SITW* was used for testing.

CN-Celeb: The entire dataset were used for testing. It involves $130,106$ utterances from $1,000$ speakers. In detail, we randomly selected 5 utterances of each speaker for speaker enrollment, and all of these rest utterances are used for evaluation. Note that the enrollment set only contains 975 of $1,000$ speakers, as there are 25 speakers containing less than 5 utterances.

3 Setting

3.1 Preprocessing

The 40-dimensional Fbanks are used as the input feature. Data augmentation was applied, where the MUSAN corpus [**?**] was used to generate noisy utterances and the room impulse responses (RIRS) corpus [**?**] was used to generate reverberant utterances. The input segment has 200-400 frames.

3.2 Network architecture

For the x-vector system, there are two DNN architectures, TDNN topology and and ResNet-34 topology. For TDNN topology, it follows the Kaldi SITW recipe [**?**]. For ResNet-34 topology, it follows the champion model in *VoxCeleb Speaker Recognition Challenge* 2019 [**?**]. The penultimate hidden layer (*segment*6) are regarded as the x-vector embedding layer. More details are shown in Table 1 and Table 2.

Table 1 The TDNN topology. The N in the softmax layer corresponds to the number of training speakers.

Layer	Layer context	Output
Frame1	[t-2, t-1, t, t+1, t+2]	512
Frame2	[t-2, t-1, t, t+1, t+2]	512
Frame3	[t-3, t-2, t-1, t, t+1, t+2, t+3]	512
Frame4	[t]	512
Frame5	[t]	512
Pooling	[0, T]	-
Segment6	[0, T]	512
Segment7	[0, T]	512
Classifier	[0, T]	N

Table 2 The ResNet-34 topology. The N in the softmax layer corresponds to the number of training speakers.

Layer	Structure	Output
Conv2D-1	3×3, Stride 1	512
ResNetBlock-1	$[3 \times 3 \times 32, 3 \times 3 \times 32] \times 3$, Stride 1	$40 \times L \times 32$
ResNetBlock-2	$[3 \times 3 \times 64, 3 \times 3 \times 64] \times 3$, Stride 2	$20 \times L/2 \times 32$
ResNetBlock-3	$[3 \times 3 \times 128, 3 \times 3 \times 128] \times 3$, Stride 2	$10 \times L/4 \times 64$
ResNetBlock-4	$[3 \times 3 \times 256, 3 \times 3 \times 256] \times 3$, Stride 2	$5 \times L/8 \times 128$
Pooling	-	-
Segment6	-	512
Segment7	-	512
Classifier	-	N

3.3 Pooling strategy

Both the standard statistic pooling [**?**] and the self-attentive pooling [**?**] strategies are implemented. To further study the self-attentive pooling strategy, we conduct a individual experiment to investigate the role of the attention head. Experimental results conducted on *CN-Celeb* are shown in Table 3. For *CN-Celeb*, it seems that setting the number of head to 1 is optimal.

Table 3 Multi-head attention pooling on CN-Celeb.

Multi-head	1	2	3	4	5
EER(%)	**12.06**	12.22	12.52	12.13	12.62

3.4 Loss function

The loss functions of Softmax and ArcSoftmax [**?**] are implemented. ArcSoftmax also called 'Additive Angular Margin Softmax'. Compared with the basic Softmax, ArcSoftmax introduces angular margin, weight normalization and feature normalization. Recent studies [**?**] showed that ArcSoftmax has great advantage in the x-vector system. Although its great success, ArcSoftmax is hard to optimize. It also need to be applied with some tricks (e.g., pre-training and fine tuning).

Besides, we also conducted an extract experiment to test the optimal angular margin m for *CN-Celeb* dataset. Experimental results are shown in Table 4. It presented that m=0.10 is optimal for *CN-Celeb*. Note that the empirical value of m on *VoxCeleb* is from 0.20 to 0.25. This may attribute to the reason that the *CN-Celeb* covers more complex conditions, so the margin among different speakers should be much smaller.

Table 4 The effect of angular margin in ArcSoftmax on CN-Celeb.

Angular margin	0.05	0.10	0.15	0.20	0.25
EER(%)	12.64	**12.49**	12.77	12.64	12.78

3.5 Back-end scoring

The 512-dimensional x-vectors extracted from the penultimate hidden layer are firstly projected to 128-dimensional vectors by LDA, and then the PLDA model was employed to score the trials.

4 SOTA models

4.1 Quantitative analysis

This section presents the performance of SOTA speaker recognition methods on both *SITW* and *CN-Celeb*. Experimental results are shown in Table 5 and Figure 2.

Firstly, it can be observed that among all these SOTA speaker recognition approaches, the performance on *CN-Celeb* is far inferior to the one obtained on *SITW*. It demonstrated that *CN-Celeb* is a more challenging dataset. Secondly, for the network architecture, the ResNet-34 shows its advantage to TDNN. For the pooling strategy, the attentive pooling is better than the standard statistical pooling. For the loss function, ArcSoftmax wins Softmax.

Table 5 EER performance of SOTA speaker recognition methods on both SITW and CN-Celeb.

Network	Pooling	Loss	EER(%) on SITW	EER(%) on CN-Celeb
TDNN	Stat.	Softmax	2.433	16.07
TDNN	Att.	Softmax	2.406	15.97
TDNN	Stat.	ArcSoftmax	2.488	15.82
TDNN	Att.	ArcSoftmax	2.570	15.96
ResNet-34	Stat.	Softmax	2.406	**15.37**
ResNet-34	Att.	Softmax	2.160	15.56
ResNet-34	Stat.	ArcSoftmax	**1.958**	15.56
ResNet-34	Att.	ArcSoftmax	2.296	15.49

4.2 Qualitative analysis

The discriminative capability of the x-vectors can be visualized more clearly by projecting the feature vectors to a 2-dimensional space. We choose t-SNE [**?**] to perform the projection. The distribution of x-vectors among 10 speakers are visualized

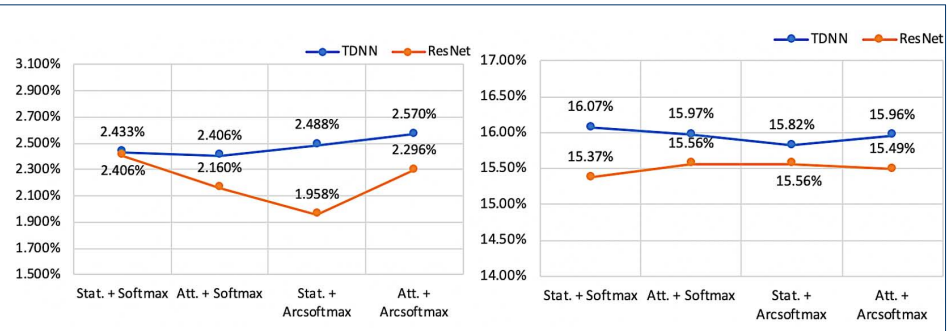

Figure 1 EER performance of SOTA speaker recognition methods on *SITW*(left) and *CN-Celeb*(right).

in Figure 2. All these x-vectors are extracted from 'ResNet-34' plus 'ArcSoftmax' models (the 7th row in Table 5).

It can be observed that there is a clear margin among different speakers, which represents that the x-vector has strong discriminative capability. However, there also exists a domain mismatch problem in each speaker space, leading to a larger within-speaker variance and limited recognition performance.

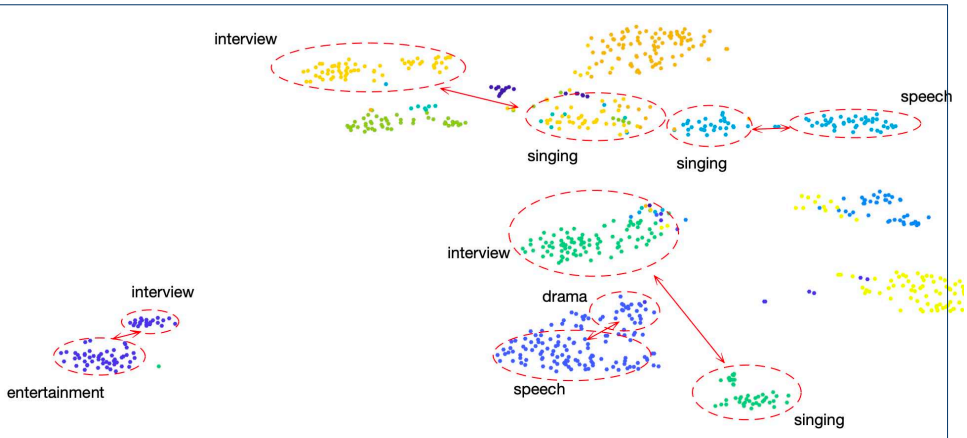

Figure 2 The distribution of x-vectors among 10 speakers are plotted by t-SNE, with each color representing a speaker.

5 Factors in CN-Celeb

In this section, we will give further analysis on *CN-Celeb*, and focus on several notable factors, including genres, duration, gender, age. All these experiments in this section are based on 'ResNet-34' plus 'ArcSoftmax' models (the 7th row in Table 5).

5.1 Genres

For *CN-Celeb*, it covers over 11 different genres, including entertainment, interview, music, play, movie, vlog, live broadcast, speech, drama, recitation and advertisement and so on. Here we made a full-cross experiment among these genres. For each genre, 5 utterances are randomly selected for speaker enrollment, and the rest are used

for test. All these cross-genre trials are pooled together to obtain the total trials. Experimental results are shown in Figure 3.

enroll \ test	01-entertain	02-interview	03-singing	04-play	05-movie	06-vlog	07-live_broa	08-speech	09-drama	10-recitation	11-advertise	total
01-entertain	16.15%	16.70%	26.89%	24.62%	20.57%	22.30%	18.71%	14.75%	21.24%	16.33%	24.68%	18.30%
02-interview	19.44%	13.27%	28.46%	20.11%	28.05%	19.13%	18.18%	14.47%	25.93%	16.41%	17.00%	17.70%
03-singing	29.04%	27.63%	25.68%	28.12%	37.17%	29.16%	25.86%	21.63%	35.05%	23.44%	33.33%	26.10%
04-play	24.63%	21.49%	27.12%	20.09%	22.78%	11.67%	24.59%	40.30%	26.27%	20.69%	18.75%	21.32%
05-movie	21.85%	20.26%	33.06%	27.02%	22.19%	20.93%	16.13%	27.50%	23.64%		33.33%	22.42%
06-vlog	29.40%	18.05%	36.06%	12.22%	31.58%	17.29%	18.54%	16.67%	22.45%			20.52%
07-live_broa	21.57%	17.55%	27.80%	23.33%	25.71%	23.38%	13.38%	17.36%	18.34%	25.00%	10.00%	17.00%
08-speech	18.32%	14.72%	23.82%	30.10%	16.67%	25.49%	20.45%	8.61%	16.67%	16.14%	16.36%	13.03%
09-drama	23.17%	20.85%	35.52%	23.23%	33.63%	23.13%	17.13%	11.85%	21.79%	43.08%	20.00%	22.61%
10-recitation	27.16%	16.79%	22.67%	26.33%		33.33%	28.01%	14.84%	46.67%	13.23%	25.00%	21.51%
11-advertise	21.16%	16.46%	50%	13.95%	20.90%		19.05%	9.90%	33.33%		13.33%	15.96%

Figure 3 Experimental results on cross-genre test conditions.

Besides, we draw the DET curves of 11 geners, respectively. As shown in Figure 4. It can be observed that the 'speech' genre obtains the best performance while the 'singing' genre get the worst. Also, we found that the performance on the genres which are often mixed by background voices (e.g., singing, movie, drama) are generally worse than that contain only standalone conversation (e.g., speech, interview, live broadcast).

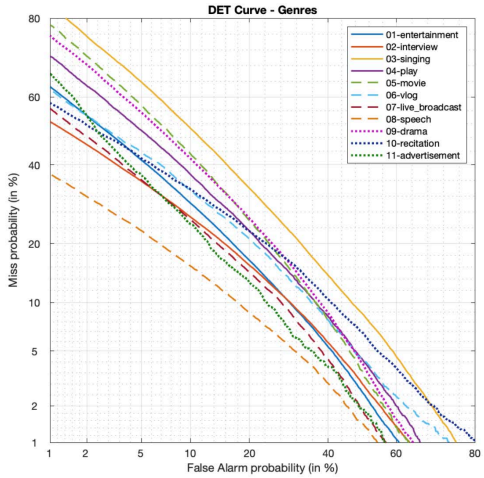

Figure 4 DET curves of 11 genres.

5.2 Speech duration

To investigate the effect of speech duration, we separated the *CN-Celeb* into 8 subset with different speech durations. Table 6 and Figure 8 show the EER performance on different speech duration. Experimental results showed that the larger speech duration, the better EER performance. It demonstrates that the short-utterance issue is a challenging task for speaker recognition. Moreover, it can be observed that when the speech duration is about $15s$, the performance seems stable.

Table 6 EER performance on different speech duration.

Duration(s)	[0,2]	[2,5]	[5,10]	[10,15]	[15,20]	[20,25]	[25,30]	[30,-]
Proportion(%)	32	30	18	8	4	2.5	1.5	4
EER(%)	21.17	12.69	8.149	6.381	5.976	6.046	6.364	5.739

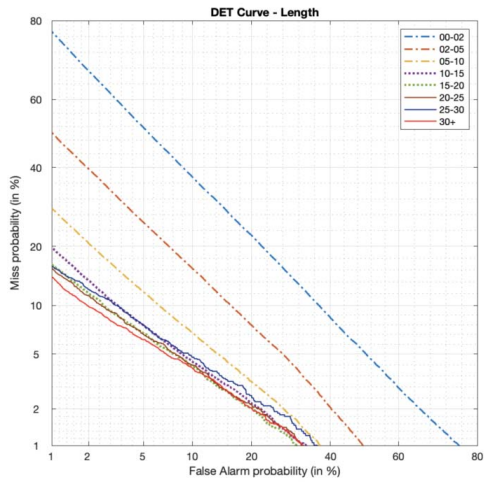

Figure 5 DET curves on different duration testsets.

5.3 Gender

The *CN-Celeb* dataset includes 423 female speakers and 577 male speakers. Each genre is divided according to gender. Experimental results are shown in Figure 3. It can be observed that the EER performance on male speakers is better than female speakers. This may attribute to the gender unbalance, as the number of male speakers is larger than that of female speakers.

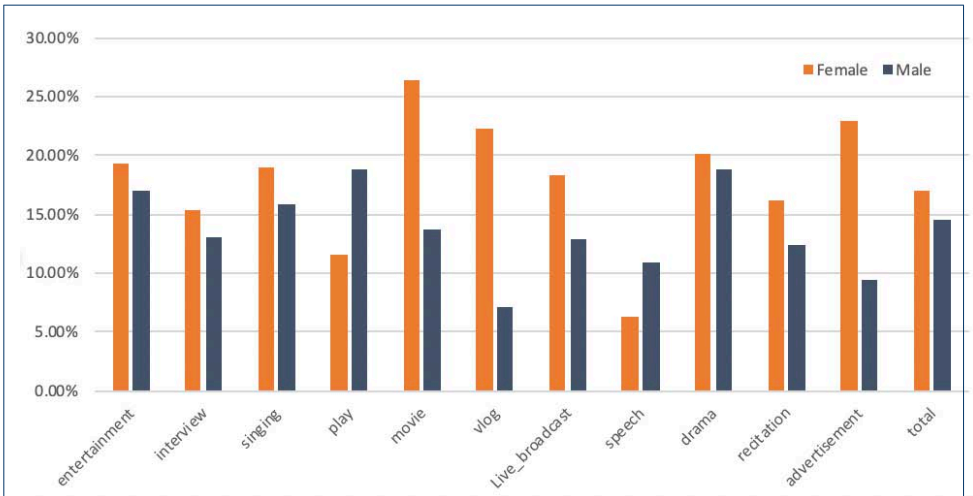

Figure 6 EER performance on male and female in each genre.

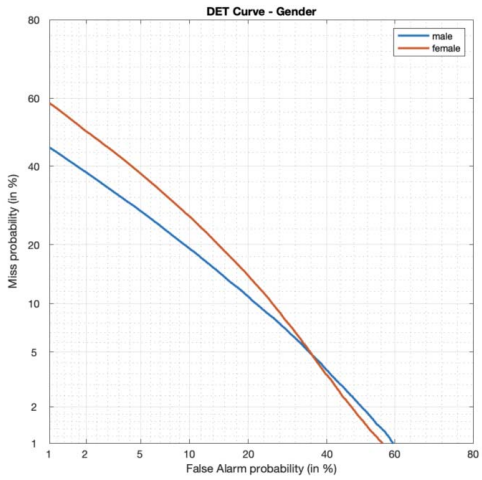

Figure 7 DET curves on male and female.

5.4 Age

Considering 80% speakers in *CN-Celeb* born from 1970 to 1990, we make an extreme comparison on two age group, in which speakers are born before 1970 or born after 1990. The data distribution over age and EER results are shown in Table 7. It can be observed that the performance on these two age groups is quite similar.

Table 7 The data distribution over age.

Age	Before 1970	After 1990
EER	14.58%	15.67%
No. of Spks	210	153
No. of utts	27,837	18,841
Avg. length	8.37s	6.61s
Ent.	15%	13%
Reci.	3%	1%
Ad.	0%	0%
Int.	44%	53%
Sin.	9%	10%
Play.	7%	1%
Mov.	1%	0%
Vlog	0%	3%
Live	0%	13%
Spe.	12%	0%
Dra.	4%	0%

6 Conclusions

The EER performance we obtained in this paper suggests that in unconstrained conditions, the performance of the current speaker recognition techniques might be much worse than it was thought.

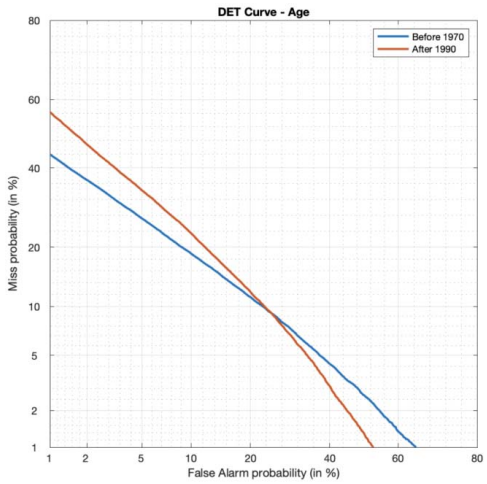

Figure 8 DET curves on 1970s and 1990s.

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